Examiner only

**5.** Thalidomide is a medical drug that caused unexpected damage to foetuses in the 1950s and 1960s. It was used for easing morning sickness in pregnant women, although it had not been fully tested.

As a result of the damage to foetuses, the drug was banned. Drug testing was also made more thorough.

Thalidomide has since been used as a treatment for bone cancer. However, its use is heavily regulated to prevent a repeat of the problems it caused last century.

With thalidomide came the widespread recognition that differences in sensitivity to drugs between species required consideration. Many scientists report that animals are good predictors of how humans will respond to drugs and that current testing procedures would have identified the dangers of thalidomide and the side effects would have been avoided.

Despite this, there is evidence that animal tests cannot predict how humans will respond to a drug. In 1990, six different drugs were trialled in humans and in animals. The results showed that animals and humans had 22 common side effects (true positives). There were 48 side effects in animals which did not occur in humans (false positives). There were also 20 side effects in humans that did not occur in animals (false negatives).

*(a)* The sensitivity of the animal tests is calculated using the formula:

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP + FN}} \times 100$$

TP = true positives

FN = false negatives

Calculate the sensitivity of the animal tests in the trial carried out in 1990. [2]

Sensitivity = ...................................... %

Examiner only

(b) Use **only** the information in the passage to answer the following questions.

(i) Evaluate the arguments **for** and **against** the use of animals in drug testing. [2]

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(ii) State how the thalidomide problem has had **one** positive outcome on the development of new drugs. [1]

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(iii) State why the ban on thalidomide was removed. [1]

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